

The Theory of Finite-Mode Bohmian–Hartree Dynamics: A Technical Monograph

1.0 Introduction: A Quantum-Classical Framework with Finite Memory

The finite-mode Bohmian-Hartree model represents a specialized yet powerful framework for analyzing quantum systems that interact with a limited set of classical degrees of freedom. It provides a mathematically consistent description of a quantum wavefunction coupled to a finite bank of harmonic oscillators, where the interaction is mediated by nonlocal Hartree-type terms. This monograph systematically deconstructs the model's mathematical structure, its physical interpretation as a system with "finite memory," and its theoretical context within the broader landscape of quantum physics.

We will explore the complete dynamics of the system, which are derived from a fundamental action principle. This derivation yields a coupled Schrödinger-ODE system that governs the co-evolution of the quantum wavefunction and the classical oscillator coordinates. A key feature of this construction is the existence of a conserved total energy, which serves as a crucial check on the physical consistency of the model. Within this framework, we define Bohmian trajectories for the quantum particle according to the standard guidance law, where the particle's velocity is determined by the phase of the wavefunction. We will establish that these trajectories are statistically consistent with the predictions of quantum mechanics through the property of $|\Psi|^2$ -equivariance, which ensures that an initial probability distribution matching the quantum density is preserved over time.

Crucially, this model is not proposed as a fundamental modification of quantum mechanics. Instead, its intended application is as an *effective* quantum-classical closure. It is designed to approximate the behavior of a quantum system interacting with a complex environment or a feedback mechanism by representing those external influences with a small, manageable set of classical modes. This monograph will detail the theory's formulation, its analytical properties, and its place as a well-posed and insightful tool for modeling specific physical regimes.

2.0 The Mathematical Formulation of the Dynamics

A rigorous mathematical foundation is essential for any physical theory to be predictive and internally consistent. This section establishes that foundation for the finite-mode Bohmian-Hartree system. We will begin by precisely defining the state space and its constituent components. From there, we will introduce the fundamental action principle from which the system's dynamics are derived. Applying the principle of stationarity to this action yields the coupled equations of motion that govern the system's evolution. Finally, we establish a critical conservation law for the total energy, which confirms the coherence of the resulting dynamical structure.

2.1 State Space and System Components

The complete state of the system at any given time is specified by a combination of quantum and classical variables, along with a set of fixed physical parameters. These components are defined as follows:

- Quantum Component:** The state of the quantum particle is described by a nonrelativistic wavefunction $\Psi(\mathbf{x}, t)$, a complex-valued function on $\mathbb{R}^d \times [0, \infty)$ in d spatial dimensions ($d \in \{1, 2, 3\}$). The wavefunction is normalized to unity, $\|\Psi(t)\|_{L^2} = 1$, and determines the quantum probability density $\rho(\mathbf{x}, t) = |\Psi(\mathbf{x}, t)|^2$.
- Classical Components:** The classical part of the system consists of a finite set of M harmonic oscillators. Their state is described by:
 - A set of real-valued coordinates $q_k(t)$, indexed by $k \in I = \{1, \dots, M\}$.
 - A corresponding set of momenta $\pi_k(t) = \dot{q}_k(t)$.
- System Parameters:** The model is defined by a set of fixed parameters:
 - $\omega_k > 0$: The natural frequencies of the classical oscillators.
 - $g_k \in \mathbb{R}$: A set of coupling constants that determine the strength of the interaction between the quantum particle and each classical mode.
 - $K_k(\mathbf{x})$: A set of even interaction kernels ($K_k(\mathbf{x}) = K_k(-\mathbf{x})$) belonging to the space $L^1(\mathbb{R}^d)$. These kernels define the spatial form of the nonlocal interaction.
- Hartree Functionals:** The interaction between the quantum and classical components is quantified by a set of Hartree functionals, $H_k[\rho]$. These functionals measure the interaction energy associated with each mode k and are defined by the double integral:

$$H_k[\rho] := \frac{1}{2} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \rho(\mathbf{x}) K_k(\mathbf{x}-\mathbf{y}) \rho(\mathbf{y}) d\mathbf{x} d\mathbf{y}$$
- This double-integral form is characteristic of mean-field theories, where each part of the quantum probability "fluid" $\rho(\mathbf{x})$ interacts with every other part $\rho(\mathbf{y})$ via the kernel $K_k(\mathbf{x}-\mathbf{y})$.

2.2 The Action Principle

The entire dynamics of the coupled system can be derived from a single action integral, S . The equations of motion arise from the principle of stationary action, which requires that the variation of S with respect to the dynamical variables (Ψ , Ψ^* , and q) is zero. The action is given by:

$$S[\Psi, \bar{\Psi}, q] = \int dt \left(\text{Re} \langle \Psi, i\hbar \partial_t \Psi \rangle - \langle \bar{\Psi}, \hat{H}_0 \Psi \rangle - \sum_{k \in I} g_k q_k(t) H_k[\rho(t)] + \sum_{k \in I} \left(\frac{1}{2} \dot{q}_k(t)^2 - \frac{1}{2} \omega_k^2 q_k(t)^2 \right) \right)$$

where $\langle f, g \rangle = \int_{\mathbb{R}^d} f^* g \, dx$ denotes the standard L^2 inner product and $\hat{H}_0 = -(\hbar^2/2m)\Delta + V_{\text{ext}}(x)$ is the quantum Hamiltonian for a single particle with external potential V_{ext} . The integrand can be deconstructed into four key parts:

1. The quantum kinetic term ($\text{Re} \langle \Psi, i\hbar \partial_t \Psi \rangle$).
2. The quantum potential energy ($\langle \Psi, \hat{H}_0 \Psi \rangle$).
3. The central interaction term ($\sum g_k q_k(t) H_k[\rho]$), which couples the classical coordinates q_k to the quantum density ρ .
4. The classical Lagrangian, containing the kinetic and potential energies of the M oscillators.

2.3 The Coupled Equations of Motion

Applying the principle of stationarity to the action S yields two sets of coupled differential equations that define the time evolution of the system.

- **The Schrödinger Equation (S)** The evolution of the wavefunction Ψ is governed by a mode-modulated Hartree equation:
 - $i\hbar \partial_t \Psi = \hat{H}_0 \Psi + \left(\sum_{k \in I} g_k q_k(t) (K_k * \rho)(x, t) \right) \Psi$, where $\rho = |\Psi|^2$
 - This is a nonlinear Schrödinger equation where the nonlocal Hartree potential, $(K_k * \rho)$, is multiplied by the time-dependent classical coordinates $q_k(t)$. Each $q_k(t)$ acts as a time-dependent **coupling coefficient** for the k -th component of the nonlinear potential, directly influencing the quantum evolution.
- **The Oscillator Equations (O)** The evolution of the classical coordinates q_k and momenta π_k is governed by a set of driven harmonic oscillator equations:
 - $\dot{q}_k = \pi_k$
 - $\dot{\pi}_k = -\omega_k^2 q_k - g_k H_k[\rho(t)]$
 - In this formulation, the quantum state acts back on the classical system. The Hartree functional $H_k[\rho]$ acts as a state-dependent **driving force** or **source term** in the oscillator's equation of motion. The system thus constitutes a feedback loop where the quantum and classical components mutually influence each other's evolution.

2.4 The Conserved Total Energy

A fundamental property of this dynamical system is the conservation of total energy. The total energy $E(t)$ is the sum of the quantum energy, the classical energy, and the interaction energy:

$$E(t) = \int_{\mathbb{R}^d} \left(\frac{\hbar^2}{2m} |\nabla \Psi|^2 + V_{\text{ext}} |\Psi|^2 \right) dx + \sum_{k \in I} \left(\frac{1}{2} p_k^2 + \frac{1}{2} \omega_k^2 q_k^2 \right) + \sum_{k \in I} g_k q_k H_k[\rho]$$

Proposition 1 (Energy Conservation): For sufficiently regular solutions of the coupled equations (S)–(O), the total energy is conserved, i.e., $dE(t)/dt = 0$.

The existence of this conservation law is a powerful indicator of the internal consistency of the model. It provides a physical constraint that any valid solution must satisfy and serves as an essential diagnostic tool in numerical simulations. This completes the formal description of the system's state evolution. The next section addresses the motion of the particle itself within this evolving quantum-classical landscape.

3.0 The Bohmian Framework: Trajectories and Equivariance

Having defined the evolution of the wavefunction Ψ and the classical modes, the next logical step is to define the motion of the quantum particle itself. In the de Broglie-Bohm interpretation of quantum mechanics, the wavefunction does not just provide statistical predictions; it actively guides the motion of a point particle along a deterministic trajectory. The cornerstone of this interpretation's consistency is the property of *equivariance*, which guarantees that the statistical predictions derived from these trajectories align perfectly with those of standard quantum mechanics.

3.1 The Guidance Law and Bohmian Trajectories

The motion of the particle is dictated by a velocity field $v\Psi(x, t)$ that is generated directly from the wavefunction Ψ . This velocity field is defined in terms of the probability current $j = (\hbar/m) \text{Im}(\Psi^* \nabla \Psi)$ and the probability density $\rho = |\Psi|^2$:

$$v^\Psi(x, t) = \frac{j(x, t)}{\rho(x, t)} = \frac{\hbar}{m} \text{Im} \left(\frac{\nabla \Psi}{\Psi} \right)(x, t)$$

This definition is valid at all points x where $\Psi(x, t) \neq 0$. The Bohmian trajectory, denoted X_t , is the integral curve of this velocity field. It is the solution to the first-order ordinary differential equation known as the **guidance equation**:

$$(B) \quad \dot{X}_t = v^\Psi(X_t, t)$$

This equation postulates a deterministic evolution for the particle's position. The particle's velocity at any point in space and time is not an independent variable but is completely determined by the local phase of the quantum wavefunction. The particle is thus "piloted" by the wave.

3.2 The Continuity Equation and Statistical Equivariance

The mathematical link between the evolution of the probability density ρ and the velocity field $v\Psi$ is the continuity equation.

Proposition 2 (Continuity Equation): If Ψ is a sufficiently regular solution of the Schrödinger equation (S), then the following equation holds: $\partial_t \rho + \nabla \cdot (\rho v\Psi) = 0$.

This equation shows that the probability density ρ evolves as a conserved fluid with a flow velocity $v\Psi$. This fluid-like behavior is the foundation for the central consistency result of Bohmian mechanics: $|\Psi|^2$ -equivariance.

Theorem 1 ($|\Psi|^2$ -equivariance): Assume that for a time interval $t \in [0, T]$, the velocity field $v\Psi(\cdot, t)$ generates a measurable flow on $\mathbb{R}^d \setminus Z_t$, where $Z_t = \{x : \Psi(x, t) = 0\}$ is the set of the wavefunction's nodes. Assume also that trajectories avoid these nodes. If the initial position of the particle, X_0 , is a random variable with probability density $\rho(\cdot, 0) = |\Psi(\cdot, 0)|^2$, then the position at any later time, X_t , will have the probability density $\rho(\cdot, t) = |\Psi(\cdot, t)|^2$.

The implication of this theorem is profound. It guarantees that if we assume an initial state of "quantum equilibrium" where the particle distribution matches the quantum probability density, this equilibrium is preserved by the dynamics. Consequently, for any observable related to position, the statistical predictions of the Bohmian model are identical to those of standard quantum mechanics. Equivariance ensures that the hidden trajectories, while deterministic, reproduce the statistical results of the Born rule. Having established the conceptual consistency of these trajectories, we must now address whether the underlying equations that generate them are mathematically well-behaved.

4.0 Analytical Foundations: Well-Posedness of the System

For a physical theory to be predictive, its governing equations must be well-posed. This mathematical property ensures that for a given set of physically reasonable initial conditions, a unique solution exists, is stable, and persists for some period of time. Well-posedness is a prerequisite for the model to be computationally tractable and to make unambiguous physical

predictions. This section addresses the conditions under which the finite-mode Bohmian-Hartree system meets this critical requirement.

4.1 Mathematical Characterization

The complete system, defined by equations (S) and (O), constitutes a coupled partial differential equation-ordinary differential equation (PDE-ODE) system. Its components can be characterized as follows:

- The Schrödinger equation (S) is a **semilinear/nonlinear Schrödinger equation** of the Hartree-type. Its nonlinearity arises from the $(Kk * |\Psi|^2)\Psi$ term, and it is distinguished by the presence of time-dependent coefficients, $qk(t)$, which modulate the strength of this nonlinearity.
- The oscillator equations (O) form a **smooth, finite-dimensional ordinary differential equation system**. The driving force depends on the solution of the PDE, but for any given wavefunction Ψ , the equations for q and π are well-behaved.

4.2 Conditions for Local Existence and Uniqueness

The existence of local-in-time solutions can be guaranteed under a set of minimal, checkable hypotheses on the initial data and system parameters.

Assumption 1 (Baseline Regularity):

- The external potential is sufficiently smooth: $V_{\text{ext}} \in W^{2,\infty}(\mathbb{R}^d)$.
- The interaction kernels are integrable and even: $Kk \in L^1(\mathbb{R}^d)$.
- The initial wavefunction is sufficiently regular: $\Psi_0 \in H^2(\mathbb{R}^d)$ with $\|\Psi_0\|_{L^2} = 1$.
- The initial classical state is well-defined: $qk(0), \pi k(0) \in \mathbb{R}$.

Under these standard conditions, the existence of a unique local solution is guaranteed.

Theorem 2 (Local Well-Posedness): Under the baseline regularity conditions of Assumption 1, there exists a time $T > 0$ and a unique solution to the coupled system (S)–(O) such that the wavefunction exists in the space $\Psi \in C([0, T]; H^2)$, and the classical trajectory exists in $(q, \pi) \in C^1([0, T]; \mathbb{R}^{2M})$. Moreover, the L^2 norm of the wavefunction is conserved for all $t \in [0, T]$.

This theorem ensures that the model is well-defined and predictive, at least for short times. The question of global well-posedness (whether solutions exist for all $t > 0$) is more complex and depends on more specific properties of the system, such as the spatial dimension and the functional class of the kernels. The broader mathematical literature on Hartree-type equations provides numerous theorems establishing global results under more restrictive conditions, while rigorous work has established global existence for Bohmian trajectories in the linear case for typical initial data [6].

5.0 Physical Interpretation and Theoretical Context

A model's utility is defined not only by its mathematical consistency but also by its physical interpretation and its relationship to established theories. A well-posed set of equations can represent anything from a fundamental law of nature to a convenient approximation for a specific physical regime. This section positions the finite-mode Bohmian-Hartree system correctly as an effective model and evaluates its scope and limitations in light of both theoretical principles and experimental constraints.

5.1 An Effective Theory of Quantum-Classical Interaction

The primary interpretation of this model is as an **effective theory**, not a fundamental one. The finite bank of classical oscillators is not intended to represent a true, fundamental classical sector of reality coexisting with the quantum world. Instead, it serves as a simplified, coarse-grained representation of a more complex quantum environment or a classical feedback layer.

In this view, the oscillators provide a "finite memory" closure for an open quantum system. The mechanism for this memory is direct: the oscillator's equation of motion implies that its position $q_k(t)$ depends on the integral of the force $H_k[p]$ over past times, thereby encoding a memory of the quantum state's history. This memory then feeds back into the future evolution of Ψ . This makes the model suitable for describing semiclassical feedback loops or systems where a few environmental modes are dominant.

5.2 Relationship to Established Physical Models

The structure of the finite-mode Bohmian-Hartree system bears a strong resemblance to several well-known physical models, which helps to place it in a broader context.

- **Caldeira–Leggett Model:** This model is a cornerstone of open quantum systems theory, describing a quantum system coupled to a bath of infinitely many harmonic oscillators [7]. In the limit where the number of modes M becomes very large and their frequencies ω_k and couplings g_k are chosen according to a suitable spectral density, the finite-mode model's feedback mechanism can reproduce the dissipative and stochastic forces characteristic of quantum Brownian motion, thereby providing a bridge between a simple feedback system and a full thermal bath.
- **Mean-Field and Hartree Models:** The nonlocal term $(K * \rho)\Psi$ is structurally identical to that found in self-consistent field theories like the Hartree equation, which are used to approximate the behavior of many-body quantum systems. The finite-mode model extends this structure by allowing the strength of the self-interaction to be dynamically modulated.

- **Schrödinger–Newton Equation:** This well-known equation is a specific special case of the model [9]. It arises by setting $d=3$, having a single mode ($M=1$), and choosing the kernel to be the gravitational potential, $K(x) = -Gm^2 / |x|$.
- **Electron–Phonon Coupling:** The model is conceptually analogous to theories of polaron physics, where an electron moving through a crystal lattice interacts with collective vibrational modes (phonons) [8]. In this analogy, the oscillators q_k represent the phonon modes, which modulate the effective interactions experienced by the electron.

5.3 Scope and Empirical Constraints on Nonlinearity

If one were to propose this model as a fundamental replacement for linear quantum mechanics, it would face severe empirical and theoretical challenges. Any such nonlinear modification is strongly constrained.

- **Experimental Tests:** Following the theoretical framework for general nonlinear quantum theories developed by Weinberg [1, 2], precision experimental tests were conducted to search for evidence of such effects. The experiments, notably by Bollinger et al. using trapped ion spectroscopy, yielded null results and placed extremely tight bounds on the potential magnitude of any fundamental nonlinearity [3].
- **Theoretical Prohibitions:** Beyond experimental limits, fundamental nonlinear modifications to the Schrödinger equation are known to have profound theoretical consequences. As shown by Polchinski [4] and through arguments of the Gisin type [5], generic nonlinear evolutions, when combined with the principles of quantum entanglement, can lead to unphysical phenomena such as the ability to send signals faster than the speed of light.

These powerful constraints firmly place the finite-mode Bohmian-Hartree model in the domain of effective, regime-specific descriptions. Its value lies not in challenging quantum mechanics, but in providing a consistent and computable tool for modeling complex quantum-classical feedback systems where a full quantum description is intractable.

6.0 A Computable Example and Advanced Topics

To solidify the theoretical concepts discussed, it is useful to consider a simplified, concrete instance of the model that is amenable to direct numerical simulation. This allows for the exploration of the system's dynamics in practice. Furthermore, we can briefly touch upon more advanced theoretical considerations that arise when applying the Bohmian framework to non-trivial configuration spaces.

6.1 A Minimal 1D Toy Model

A minimal, computable version of the model can be constructed in one spatial dimension ($d=1$) with a single classical mode ($M=1$). By choosing a simple exponential kernel, $K(x) = e^{-|x|/\ell}$ where $\ell > 0$ is a characteristic length scale, the full system reduces to a manageable set of equations:

- **Schrödinger Equation:**
- $i\hbar\partial_t \Psi = -\frac{\hbar^2}{2m} \partial_{xx} \Psi + V_{\text{ext}} \Psi + g q(t) (K * |\Psi|^2) \Psi$
- **Oscillator Equation:**
- $\ddot{q} + \omega^2 q = -g H[\rho]$

The corresponding **Bohmian trajectory equation** in 1D is given by:

$$\dot{X}_t = \frac{\hbar}{m} \text{Im} \left(\frac{\partial \Psi}{\partial x} \right) (X_t, t)$$

This "toy model" encapsulates all the essential features of the full theory—quantum-classical feedback, nonlinearity, and Bohmian guidance—in a form that is ideal for numerical investigation. A common computational approach would be to use a split-step Fast Fourier Transform (FFT) method to evolve the wavefunction Ψ , coupled with a standard ODE integrator (like Runge-Kutta) to evolve the classical coordinate q .

6.2 Topological Considerations in Configuration Space

When the configuration space of the quantum particle is not simply connected (for example, $\mathbb{R}^2$ with a region removed), the Bohmian velocity field can exhibit interesting topological properties. For a closed loop γ that does not pass through a node of the wavefunction, one can define the **circulation** of the velocity field:

$$\Gamma(t) = \oint_{\gamma} v(\cdot, t) \cdot d\ell$$

This quantity has two crucial properties:

1. **Quantization:** The circulation is quantized in integer multiples of $2\pi\hbar/m$.
2. **Invariance:** This quantized value remains constant over time, $d\Gamma/dt = 0$, as long as a node of the wavefunction does not cross the loop γ .

The quantization of circulation is a direct consequence of the mathematical structure of the velocity field and the single-valuedness of the wavefunction. Writing the wavefunction in polar form, $\Psi = |\Psi| e^{iS/\hbar}$, gives the velocity field as $v = \nabla S/m$. The requirement that Ψ be single-valued means that the change in its phase S around any closed loop must be an integer multiple of $2\pi\hbar$. This directly implies the quantization of $\oint v \cdot d\ell$.

7.0 Conclusion

This monograph has detailed the complete mathematical and physical structure of the finite-mode Bohmian-Hartree dynamics. We have shown that the model is derived from a coherent action principle, leading to a coupled system of equations for a quantum wavefunction and a finite set of classical oscillators. This system respects a fundamental conservation law for total energy, reinforcing its internal consistency.

Furthermore, we established that a Bohmian interpretation can be consistently applied to this framework. The particle trajectories, governed by the phase of the wavefunction, reproduce the statistical predictions of the quantum density through the crucial property of $|\Psi|^2$ -equivariance. The underlying equations of motion are mathematically well-posed under standard regularity conditions, ensuring the model is both predictive and suitable for numerical analysis.

Ultimately, the finite-mode Bohmian-Hartree system should be understood not as a proposal for new fundamental physics, but as a well-posed, computationally accessible, and physically insightful **effective theory**. It provides a robust framework for modeling quantum systems subject to feedback from a simplified, coarse-grained environment, making it a valuable tool for studying complex quantum-classical interactions with finite memory.

8.0 References

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